

Christabel Ngwenya

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Summary

A versatile recent graduate in Computer Systems Engineering with skills in network administration ,software development and hardware systems. I have experience building interactive websites with React.js, Node.js, and Laravel, and developing embedded systems with ESP32 micro-controllers and sensor networks. Proven ability to manage the entire project from start to finish, from designing a circuit and building a prototype to launching a complete web application with real-time data dashboards .I am a highly motivated and detail oriented with a history of creating successful projects such as a hackathon winning safe sphere application. I am eager to use my technical skills, adaptability, and attention to detail in a challenging role that supports learning, teamwork, and professional growth.

Work Experience

IT Intern

City Of Bulawayo , Zimbabwe, Bulawayo

August 2023 –31 July 2024

- Repaired and maintained desktop computers ,printers and peripherals.
- Participated in development and testing of internal software projects.
- Maintained and updated council websites using C sharp framework.
- Assisted in analyzing existing systems and recommending improvements.
- Supported local area network(LAN) setup and troubleshooting.
- Collaborated with teams to ensure smooth operation of the IT infrastructure.

Education

Bachelor of Science Honours Degree in Computer Systems Engineering

Class Division: 2.1

Midlands State University, Gweru

Graduated: 2025

Zimsec (Advanced Level)

Four A'Level Passes in Pure Mathematics, Physics, Chemistry and Mechanical Mathematics

Sikhulile High School, Bulawayo

Graduated: 2020

Zimsec (Ordinary Level)

Nine O'Level Passes

Sikhulile High School, Bulawayo

Graduated: 2018

Skills

Programming :Python(Data Analysis),Javascript(Reactjs,Nodejs,Express.js)

Web Development: HTML, CSS, PHP, Javascript

Data Systems: Database Management : MySQL, PostgreSQL,ELK Stackmonitoring ,IOT security

Security and Networking: **Network** configuration and troubleshooting, cybersecurity fudamentals,VPN and Firewall Management,Cisco Networking Basics, Cybersecurity Foundations, LAN/WAN Concepts

Devops and CI/CD pipelines: Git(Version control),Github Actions(CI/CD pipelines)

Soft skills:Problem solving,Team collaboration

Agile and Collaboration: Scrum methodology ,Jira for project management

AI & Data Science: Machine Learning, Deep Learning (YOLOv8), Computer Vision, Data Analysis, Reinforcement Learning, Predictive Modeling

Networking tools: Cisco packet tracer

Libraries: pandas, NumPy, Matplotlib, Pillow, Tensorflow, chart.js

Documenting and Reporting: Microsoft packagess (word,excel,PowerPoint)

Certifications 2024

- Introduction to Data Science by Cisco Networking Academy
- Networking basics Cisco Networking Academy
- Cyber Security by Oxford Home Study Centre
- Cybersecurity Foudations by Linkedin learning
- Introduction to Internet of Things by Cisco Networking Academy
- Developing Machine Learning Solutions by AWS
- Introduction to Machine Learning by AWS
- Fundamentals of Machine Learning and Artificial Intelligence by AWS
- Deep Dive with Security: AWS Identity and Access Management (IAM) by AWS
- Computer packages certification by Hexco(2018)

Projects

Portfolio Website | React, CSS, GitHub Pages, GitHub Actions (<https://christabelngwenya.github.io>)

- Designed and deployed a responsive single-page application showcasing my skills and projects I have worked on .
- Demonstrated proficiency in modern front-end development, version control with Git and implementing CI/CD pipelines for automated testing and deployment.

AI-Powered Smart Traffic Management System | Python, YOLOv8, Open CV, Flask, Reinforcement Learning

- Researched and developed a comprehensive AI model to optimize traffic flow simulating real-time adaptive signal control for Zimbabwean intersections in big cities like Harare.
- Implemented computer vision with YOLOv8 for accurate vehicle detection and counting and designed a reinforcement learning algorithm for dynamic signal timing.

- Built a functional Flask web dashboard for real-time monitoring and performance reporting, providing a scalable solution to reduce urban congestion.

Safe Sphere – IoT-Based Safety & Wellness Platform | React, Node.js, PostgreSQL, IoT Hardware (Microcontrollers)

- Worked as a developer on a team that won 1st Place (\$1000 USD) at the Pada Platform hackathon in Harare (March 2024). Our project was a comprehensive system designed to improve female student safety and mental health at universities in Zimbabwe. My main responsibility was developing the full-stack application. Key features included:
 - i. Secure, anonymous forms for reporting incidents and misconduct.
 - ii. A booking system for counseling services and an interactive mental health self-assessment tool.
 - iii. A "Virtual Walk Companion" feature that allows users to safely share their location in real-time with friends or security.
 - iv. I also helped engineer the hardware for a physical panic button. This device, built with a microcontroller, sends the user's precise GPS location directly to campus security through the web app when activated.
 - v. A key focus throughout the entire design process was ensuring user privacy, ease of use, and emotional safety.

Digital Certificate Issuance System | PHP, Laravel, MySQL

- Designed and developed a centralized web-based platform for city councils to digitally manage and issue business shop licenses.
- The system automates the entire process thereby storing shop owner details tracking payments, and generating printable verifiable certificate effectively replacing the error-prone handwritten methods which is currently being used and also reducing fraud.

Smart Water Management System | IoT, Full-Stack Development, Embedded Systems

- Researched, designed and built a complete IoT system to help reduce water loss in urban areas, this system was able to detect leaks in under 5 seconds a process that used to take weeks.
- Created the physical hardware for the system using ESP32 micro-controllers and water flow sensors, I programmed the ESP32 with the Arduino IDE to collect real time data and automatically control a water valve.
- **Developed two web applications:** one for system administrators and one for the public. The admin dashboard shows real-time data on interactive graphs and a map, and allows for remote control of the hardware. The public website lets community members report water leaks and track the repair status.
- Designed and programmed the core logic for detecting leaks, when a leak is detected the system automatically sounds an alarm and displays on the web dashboard the leak alert in red. An email is sent automatically to the admin team when a leak report is submitted.
- Technologies used: Arduino (C++), ESP32, IoT Sensors, React.js, Node.js, Express.js, MySQL, REST APIs, Serial Communication.

Project: Biometric Home Surveillance System

- **Objective:** Designed and prototyped an affordable, automated home security system using facial recognition to control access and enhance household safety.
 - Developed a computer vision application using Python and OpenCV for real-time face detection and identification, classifying individuals as authorized or unauthorized.
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- Developed a secure notification system by configuring a dedicated Gmail account with two-factor authentication and implementing the Gmail API to ensure reliable, secure communication.
- The system automatically captures an image of an unauthorized person and instantly sends an alert email with the photo attached directly to the homeowner, enabling immediate awareness and response.
- Technologies: Python, OpenCV, Gmail API, OAuth 2.0 Authentication, Haar Cascades.

References

Mr. V.J .Sibanda

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